

Week 8

Networking and Storage for Scalable AI Workflows in the Cloud

- Understand how to connect and structure ML resources in Azure
- Configure essential storage for persistent and scalable data workflows
- Prepare for real-world ML infrastructure challenges

Learning Objectives

- Design a virtual network to connect ML resources and manage data flow
- Add and configure storage for persistent datasets in Azure
- Test and document access across compute, storage, and databases
- Identify common industry practices for scalable, robust cloud ML pipelines

Agenda

1. Recap; Cloud computing for ML at NMTAFE
2. Virtual networks in Azure; concepts and practical setup
3. Storage options; persistent vs ephemeral storage
4. Hands-on; Mount Azure storage to a VM
5. Test connectivity; Compute, storage, database
6. Troubleshooting; Common networking and storage issues
7. Summary and next steps

Why Networking and Storage Matter

- Scalable ML workloads require seamless data transfer between resources
- Persistent storage ensures model reproducibility and reliable access to data
- Real-world production requires robust networking for security and speed
- Aligns with industry practices; MLOps, high performance computing, enterprise AI

Recap from Previous Weeks

- Explored Azure VMs for ML in Week 7; focused on VM types and provisioning
- Used PyTorch on local and cloud instances in Weeks 5 and 6
- Next logical step; connect resources and enable scalable, persistent dataflows

Industry Example

- A local software company uses Azure VMs to serve AI models for document processing
- Their production pipeline requires secure virtual networks plus mounted storage for customer files
- Reliable networking and storage reduce downtime and enable model versioning

Virtual Networks in Azure

- Virtual networks are private, isolated, logically grouped resources for secure traffic
- They allow control over communication between VMs, storage, and databases
- Key features; subnets, network security groups, private endpoints

Designing a Virtual Network; Key Steps

1. Define address space and subnets to group ML resources
2. Create a resource group for your project
3. Add your VM and select the new virtual network
4. Use network security groups to control inbound and outbound traffic
5. Plan for growth; allow room for additional compute or storage resources

Activity; Create a Virtual Network

1. Access your Azure portal and select Resource groups
2. Create a new virtual network in your group; define a unique address range and subnet
3. Attach an existing or new VM to your network
4. Note subnet and IP allocations for resource planning

Storage in Cloud ML; Ephemeral vs Persistent

- Ephemeral storage; for temporary files and fast computation, data lost after VM stops
- Persistent storage; for datasets, model checkpoints, and reproducible training
- Azure options; Blob storage, managed disks, file shares

Selecting Storage Solutions

Use Case	Recommended Storage	Example
Training large ML models	Azure Blob or File Share	Store training data, logs
Model checkpoints	Managed Disk or Blob	Save weights, resumes
Multiple VMs need shared data	Azure File Share	Mount from several VMs
Fast temporary scratch space	VM Ephemeral Disk	On-the-fly preprocessing

Activity; Add and Mount Azure Storage

1. In Azure, create a new Storage Account; select Blob or File Share as type
2. Note the access keys and connection string
3. SSH into your VM
4. Use the Azure CLI to mount the storage to your file system
 - Example; `az storage file download` or `mount -t cifs <storage-path> <mount-point>`
5. Test read and write to confirm persistent storage access

Demo; Mounting Storage in Ubuntu

```
sudo apt-get update
sudo apt-get install cifs-utils
sudo mkdir /mnt/azurefiles
sudo mount -t cifs //<storage-account>.file.core.windows.net/<share-name> /mnt/azurefiles -o vers=3.0,username=<storage-account>,password=<storage-key>,dir_mode=0777,file_mode=0777,serverino
```

- Replace placeholders with your account and storage details
- Access files via `/mnt/azurefiles` from any application or your ML code

Testing Network and Storage Access

- Verify connectivity between your VM and storage with CLI tools; use `ping`, `curl`, or `az storage` commands
- Validate permissions by uploading and downloading test files
- Document IP addresses, resource names, and connectivity status
- Troubleshoot permission or firewall issues as a team

Common Pitfalls; How to Avoid Them

- Default firewall or NSG rules block storage access; always review settings
- Wrong mount options can cause slow reads or permission errors; check documentation
- Not planning address space or subnets can limit future scaling
- Forgetting to use persistent storage risks losing critical model outputs

Industry Practices

- Enterprises adopt hybrid networks; combining public and private endpoints for efficiency and security
- Shared resource groups and consistent naming conventions reduce confusion
- All data movement and connections must be auditable for compliance (Responsible AI)

Mini Case Study; Scaling an AI Pipeline

- Government department needs to scale their training jobs to multiple VMs
- Solution; design a network with subnets for compute, storage, and management
- Use Azure File Share across VMs to store results and monitor progress
- Documented connectivity supports model monitoring and future audits

Practice Exercise

- Each student; design a virtual network for a sample ML pipeline with at least one VM and storage resource
- Mount a file share and upload a sample dataset
- Test and document access from compute to storage; present your findings

Assessment Preparation

- Prepare by documenting your resource group; virtual network; storage solution
- Practice mounting storage and troubleshooting common errors
- Be able to explain in a few sentences the role of virtual networks and persistent storage in a production pipeline
- Refer to Week 10 for project-based assessment requirements

Summary and Next Steps

- Virtual networking and persistent storage are foundational for scalable, reliable ML pipelines
- Mastering these skills supports your transition into ML engineering roles
- Next week; database and secret management for secure cloud ML pipelines

References and Further Reading

- Microsoft Learn; "Configure an Azure Virtual Network"
- Azure Documentation; "Mount Azure Files shares on Linux VMs"
- PyTorch; "Data Loading and Processing Best Practices"
- NMTAFE Course Handbook

